

- Demuestre las siguientes propiedades (sin utilizar tablas de propiedades).

i). $\mathcal{L}\{x(t-t_0)\} = e^{-st_0} x(s)$

$$\mathcal{L}\{x(t-t_0)\} = \int_0^{\infty} x(t-t_0) e^{-st} dt$$

$$u = t - t_0$$

$$t = u + t_0$$

$$t \rightarrow \infty = u \rightarrow \infty$$

$$t \rightarrow 0 = u \rightarrow -\infty$$

$$\mathcal{L}\{x(t-t_0)\} = e^{-st_0} \int_{-\infty}^{\infty} x(u) e^{-su} du = e^{-st_0} x(s)$$

ii). $\mathcal{L}\{x(at)\} = \frac{1}{|a|} x(s/a)$

$$\mathcal{L}\{x(at)\} = \int_{-\infty}^{\infty} x(at) e^{-st} dt$$

si $a > 0$

$$t \rightarrow +\infty$$

$$t \rightarrow -\infty$$

$$u \rightarrow +\infty$$

$$u \rightarrow -\infty$$

$$\frac{1}{a} \int_{-\infty}^{\infty} x(u) e^{-su/a} du = \frac{1}{a} x(s/a)$$

si $a < 0$

$$(-1) \frac{1}{a} \int_{\infty}^{-\infty} x(u) e^{-su/a} du = -\frac{1}{a} x(s/a)$$

$$\mathcal{L}\{x(at)\} = \frac{1}{|a|} x(s/a)$$

iii). $\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = sX(s)$

$$\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = \int_{-\infty}^{\infty} x'(t) e^{-st} dt$$

$$\rightarrow u = e^{-st}$$

$$du = -s e^{-st}$$

$$dv = x'(t)$$

$$v = x(t)$$

$$= e^{-st} x(t) \Big|_0^{\infty} - \int_0^{\infty} s e^{-st} x(t) dt = 0 - x(0) + s \mathcal{L}\{x(t)\}$$

$$\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = sX(s) - x(0)$$

$$\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = sX(s)$$

$$\text{iv). } \mathcal{L}\{x(t) * y(t)\} = X(s) Y(s)$$

$$\mathcal{L}\{x(t) * y(t)\} = \int_{-\infty}^{\infty} (x(t) * y(t)) e^{-st} dt$$

$$= \int_{-\infty}^{\infty} x(\tau) y(\tau - t) d\tau = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\tau) y(\tau - t) d\tau e^{-st} dt$$

$$= \int_{-\infty}^{\infty} x(\tau) \int_{-\infty}^{\infty} y(\tau - t) e^{-st} dt d\tau \rightarrow \begin{array}{l} u = \tau - t \\ du = -dt \end{array} \quad t = \tau + u$$

$$= \int_{-\infty}^{\infty} x(\tau) (1) \int_{-\infty}^{\infty} y(u) e^{-s(\tau + u)} dt d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau) e^{-s\tau} d\tau \int_{-\infty}^{\infty} y(u) e^{-su} du = X(s) Y(s)$$